Abhishek Kumar Pathak

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PROFESSIONAL SUMMARY

Mobile Engineering Leader with **11 years of experience**, currently working as a **Lead Android Developer at UBS**, delivering secure digital banking solutions across **Circle One, Investments, Authentication, Accounts and Cards, Home Screen, Navigation, and Video Content journeys**. Lead end-to-end mobile feature delivery across **Android and iOS ecosystems**, covering **mobile architecture, Kotlin, Swift, Jetpack Compose, React Native, Kotlin Multiplatform (KMP), API-driven development, security, unit testing, code reviews, release readiness, performance optimization, and production incident resolution**. Apply **Clean Architecture, MVVM/MVI, modularization, CI/CD, DevOps, Cloud, and AI-assisted engineering practices** to improve scalability, code quality, developer productivity, testing, and delivery efficiency. Explore and integrate emerging AI technologies, including **LLMs, Retrieval-Augmented Generation (RAG), Agentic AI, AI-powered automation, and intelligent mobile experiences**. Serve as a **Line Manager for a team of seven engineers**, providing technical mentoring, architectural guidance, career development, delivery oversight, and engineering best practices while collaborating closely with **Product, UX/UI, Backend, iOS, QA, Security, Cloud, and DevOps teams**.

TECHNICAL SKILLS

Languages: Kotlin, Java, XML, SQL, JavaScript basics, Python basics, SwiftUI exposure

Android Platform & UI: Android SDK, Android Studio, AndroidX, Jetpack Compose, XML Views, Material Design 3, View Binding, Data Binding, Activity/Fragment lifecycle, ViewModel, LiveData, Navigation Component, RecyclerView, ConstraintLayout, Paging 3, CameraX, WorkManager, Services, Media3/ExoPlayer, FCM, Play Console, Data Structure

Architecture & State: Clean Architecture, MVVM, MVI, MVP, SOLID, Repository pattern, modularization, multi-module applications, reusable UI/domain/data components, offline-first design, state management

Concurrency & Data: Kotlin Coroutines, Flow, StateFlow, SharedFlow, structured concurrency, lifecycle-aware collection, RxJava, Room, SQLite, DataStore, SharedPreferences, local persistence, caching, synchronization, migrations

Networking & APIs: REST APIs, Retrofit, OkHttp, GraphQL, Apollo Kotlin exposure, Ktor exposure, JSON, Gson/Moshi exposure, OAuth/JWT, API contracts, session/token handling, secure communication, resilient error handling

Dependency Injection: Hilt, Dagger, Dagger 2, Dagger-Hilt, Koin exposure, dependency isolation, testable component design

Testing & Quality: JUnit, Coroutines Test exposure, MockK, Mockito, Robolectric, Espresso, Compose UI Testing, unit testing, UI testing, integration testing, regression testing, TDD concepts, PR reviews, code quality

Firebase & Product Services: Firebase Cloud Messaging, Crashlytics, Firebase Authentication, Firestore/Realtime Database, Analytics and Remote Config exposure, Google Maps and ML Kit

Performance, Security & Delivery: Firebase Crashlytics, ANR/crash analysis, Android Studio Profiler, Memory Profiler, Layout Inspector, Logcat, app-size optimization, R8/ProGuard, secure storage, Android Keystore, encrypted local storage and certificate-pinning concepts, biometric-authentication awareness, OWASP-aligned reviews, Git, GitFlow, Jenkins, GitHub Actions, GitLab CI concepts, Gradle, build variants/product flavors, Jira, Confluence, Agile/Scrum

PROFESSIONAL EXPERIENCE

Associate Director | Lead Android Engineer
UBS | Doha, Qatar | Oct 2023 – Present

Apps / Platforms: UBS Mobile Banking, UBS key4, Circle One, UBS TWINT

Core Technologies: Kotlin, Java, Jetpack Compose, Compose Multiplatform, Android SDK, MVI, Clean Architecture, Kotlin Flows, Dagger Hilt, Retrofit, GraphQL, Room Database, Media3/ExoPlayer, Firebase Crashlytics, JUnit, MockK, Espresso, Compose UI Testing, Gradle, R8/ProGuard, Git, CI/CD

- Lead design and development of customer-facing Android capabilities for **UBS Mobile Banking and Circle One**, covering **authentication, investments, accounts, cards, home dashboard, navigation, customer communication, content and video/media journeys** using **Kotlin and SwiftUI.**
- Build scalable **Jetpack Compose** features using **MVI, unidirectional data flow, state hoisting, ViewModel, Coroutines and Flow**, creating lifecycle-aware, predictable and testable banking experiences.
- Develop secure **authentication and login modules**, including session management, API-driven validation, loading/error handling, recovery states and privacy-aware customer flows, contributing to a **20% year-over-year improvement in successful login completion**.
- Deliver **Investment and Portfolio journeys**, enabling customers to access investment-related information and content through reusable Compose UI components, API integrations and modular domain/data layers.
- Build and enhance **Accounts & Cards modules**, including account/card information, customer actions, API-bound state management, data validation, error handling and secure session-aware experiences.
- Develop reusable **Home Screen, Navigation and Top-Bar components** with **AndroidX Navigation and Jetpack Compose**, supporting consistent customer journeys across multiple banking modules.
- Architect reusable **UI, domain and data layers** using **Clean Architecture, MVI, modular/multi-module architecture, Hilt/Dagger, Coroutines/Flow, Retrofit/GraphQL and Room/SQLite**, improving maintainability, separation of concerns and testability.
- Designed and developed a reusable **AndroidX Media3 / ExoPlayer library** supporting lifecycle-aware video playback, autoplay, inline/full-screen modes, playback state management, analytics callbacks and error handling; integrated across **Circle One, CDM and other banking journeys**.
- Develop comprehensive **unit, UI and integration tests** using **JUnit, MockK, Mockito, Espresso and Compose UI Testing**, covering authentication, account, API, navigation and reusable UI components.
- Reduced production **crash rate by 30%** through **Firebase Crashlytics** analysis, lifecycle-safe implementation, coroutine/threading safeguards, defensive API parsing, null-safety improvements and regression testing.
- Reduced **Android App Bundle size by 18%** through dependency cleanup, resource optimization, asset optimization, modularization, **R8/ProGuard tuning** and Gradle configuration improvements.
- Apply **AI-assisted engineering and modern developer tooling** for code exploration, debugging, refactoring, technical documentation and test-case generation, while validating outputs through code review, automated testing, security controls and engineering standards.
- Work with **cutting-edge Android technologies, modern mobile architecture patterns and AI-powered engineering tools** to continuously improve development productivity, code quality, performance, test coverage and delivery efficiency.
- Own Android **release readiness and production support**, including PR/code reviews, Gradle configuration, regression validation, rollout coordination, CI/CD checks, production monitoring and post-release issue resolution.
- Diagnose complex production issues using **Crashlytics, application logs, analytics and engineering diagnostics**, converting findings into reproducible fixes, automated regression scenarios and preventive engineering improvements.
- Partner with **Product, UX/UI, Backend, iOS, QA, Security, Cloud, Analytics and Release teams** to translate business and regulatory requirements into secure, accessible and customer-focused mobile experiences.
- Serve as **Line Manager and technical mentor for a 7-member engineering team**, providing technical guidance, architecture support, PR/code reviews, engineering best practices, career development, delivery oversight and knowledge sharing.
- Contribute to **technical design, architecture decisions, API contracts, code quality standards, sprint planning, release planning and production incident management**, ensuring reliable delivery of enterprise-scale banking applications.

Android Lead – Digital Banking
Wipro | Client: Lloyds Banking Group | London, United Kingdom | Aug 2020 – Sep 2023

Apps: Lloyds Mobile Banking, Bank of Scotland Mobile App, Halifax Mobile Banking, MBNA Mobile App

Core Technologies: Kotlin, Java, Android SDK, AndroidX, XML Views, View Binding, Data Binding, RecyclerView, ConstraintLayout, Navigation Component, MVVM, Clean Architecture, RxJava, Coroutines, Retrofit, OkHttp, REST APIs, Room/SQLite, Dagger 2, Android Keystore, WorkManager, Firebase Crashlytics, JUnit, Mockito, MockK, Espresso, Gradle, Git/GitFlow, Jenkins, CI/CD

- Led Android feature delivery across **four UK digital banking brands** from a shared codebase, covering feature planning, **Kotlin/Java development, code reviews, release support and production defect resolution**.
- Developed and maintained customer-facing **XML/View-based Android screens** using **RecyclerView, ConstraintLayout, View Binding and Data Binding**, ensuring reusable and consistent experiences across Lloyds, Halifax, Bank of Scotland and MBNA.
- Delivered secure **login, authentication, account, card, PIN and statement journeys**, including validation, caching, telemetry, error handling and configuration-driven behavior across multiple banking brands.
- Built and modernised banking features using **MVVM, Clean Architecture, Kotlin Coroutines, Retrofit, REST APIs** with secure backend integration, local persistence, defensive API parsing and resilient UI states.
- Implemented **Dagger 2 dependency injection** for repositories, API services and ViewModels, while using **Coroutines** for asynchronous processing, threading and lifecycle-aware data updates.
- Applied **Android Keystore, WorkManager and Firebase Crashlytics** for secure credential handling, reliable background processing and production crash monitoring.
- Managed shared-codebase releases across four banking brands using **Gradle build variants, configuration-based feature management, Git/GitFlow, Jenkins and CI/CD**.
- Improved application **maintainability, performance and reliability** through architecture reviews, reusable components, automated testing with **JUnit, MockK and Espresso**, and resolution of crashes, memory issues, slow screens, API bottlenecks and multithreading defects.

Senior Android Engineer
Nagarro | Gurgaon, India | Jun 2019 – Jul 2020

Apps: Lutron App, NedBank, Lalamove - Fast & Affordable

Core Technologies: Kotlin, Java, Android SDK, React Native, JavaScript, AndroidX, MVVM, XML Views, RecyclerView, ConstraintLayout, Coroutines/Flow, RxJava, Room/SQLite, Retrofit, REST APIs, Google Maps, Places API, Fused Location Provider, IoT integrations, custom views and Git

- Developed mobile features across **IoT, logistics and mobility products**, contributing to the **Lutron Android IoT application** and **Lalamove React Native application** using Kotlin/Java, React Native, REST APIs and reusable mobile components.
- Built and enhanced **Lutron smart-home / IoT features** supporting device discovery, control, status updates, automation-related journeys and responsive device-management experiences using **Android SDK, MVVM, RxJava/Coroutines, Retrofit and local persistence**.
- Contributed to **Lalamove logistics and delivery workflows using React Native**, including location-aware customer journeys, pickup/drop-off flows, address search, map interaction, order-related screens and API-driven mobile experiences.
- Implemented **real-time, map-based and location-aware functionality** using **Google Maps, Places API and Fused Location Provider** for live tracking, route rendering, markers, location search and logistics workflows.
- Applied **MVVM, ViewModel-driven state management, repository patterns and local caching** to separate presentation, domain and data concerns and improve maintainability and testability.
- Optimised interactive and image-heavy screens using **RecyclerView, ConstraintLayout, custom views, caching, pagination-aware rendering and profiling-led improvements** to enhance responsiveness and UI performance.

- Collaborated with **Product, Backend, QA and cross-functional engineering teams** to validate API behaviour, troubleshoot integration issues, improve mobile performance and deliver production-ready releases.

Senior Android Developer
Extramarks Education India Pvt. Ltd. | Noida, India | Feb 2017 – Jun 2019

Apps: Extramarks Learning App, Extramarks Teaching App

Core Technologies: Java, Android SDK, MVP, XML Layouts, View Binding, Data Binding, RecyclerView, ConstraintLayout, Custom Views, RxJava, REST APIs, SQLite, Firebase Cloud Messaging, JUnit and Git

- Developed Android features for **student learning and teacher-facing education platforms**, covering **content discovery, live classes, quizzes, mock tests, homework, attendance, automated evaluation, reporting and communication workflows** using Java/Kotlin, Android SDK and RxJava.
- Built scalable **XML-based UI components** with **RecyclerView, ConstraintLayout, View Binding, Data Binding and custom views**, while separating presentation and business logic through **MVP architecture**.
- Integrated **REST APIs, SQLite/local persistence and Firebase Cloud Messaging** for asynchronous data loading, offline support, notifications, validation, error handling and reliable classroom communication.
- Implemented and supported **assessment and academic workflow modules**, including quizzes, tests, homework, attendance, evaluation and performance-reporting journeys for students and teachers.
- Improved application maintainability and release quality through **JUnit testing, reusable components, API integration testing, bug fixing, performance tuning and Play Store release support**.

Android Developer
Polestar Solutions & Services | Noida, India | Jan 2016 – Jan 2017

Core Technologies: Java, Android SDK, XML Layouts, Activities, Fragments, RecyclerView, Adapter Patterns, REST APIs, JSON, SQLite, Local Storage, Git, Debugging

- Developed and maintained **enterprise Android applications** using **Java, Android SDK and XML layouts**, building Activity/Fragment-based screens, RecyclerView lists and adapter-driven UI components for business-facing modules.
- Integrated **REST APIs, JSON parsing and SQLite/local storage** for data-driven mobile workflows, implementing validation, loading/error states and reliable local data handling.
- Contributed to **application debugging, performance optimisation, unit testing and production-defect resolution**, improving stability and maintainability across Android modules.

CERTIFICATIONS

- Microsoft Certified: Azure AI Engineer Associate (AI-102), 2026
- Google AI Fundamentals, Jun 2026
- UBS Certified Engineer, 2024
- UBS AI Citizen, 2026

EDUCATION

Bachelor of Technology (B.Tech), Computer Science - ABES Engineering College, Dr. A.P.J. Abdul Kalam Technical University | Apr 2012 - May 2016